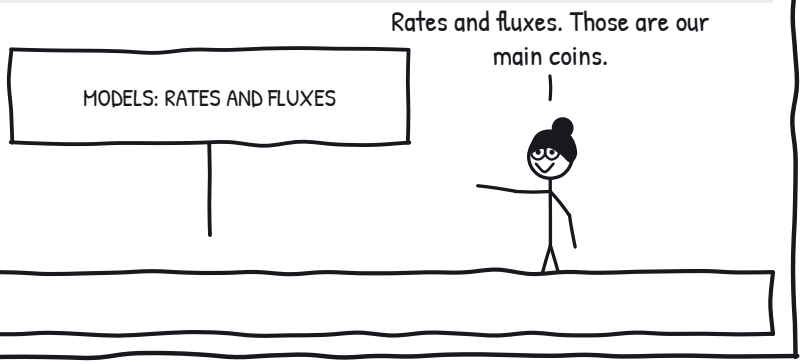
The Currency Exchange: can omics give the models what they ask for?

From "The ongoing need for rates: can physiology and omics come together to co-design the measurements needed to understand complex ocean biogeochemistry?" by Robert F Strzepek, Brook L Nunn, Lennart T Bach, John A Berges, Erica B Young, Philip W Boyd (2022)

The desk only takes one currency

Earth system models, such as those within the Coupled Model Intercomparison Project (CMIP), work in currencies that include physiological rates and the biogeochemical fluxes of bioactive elements.



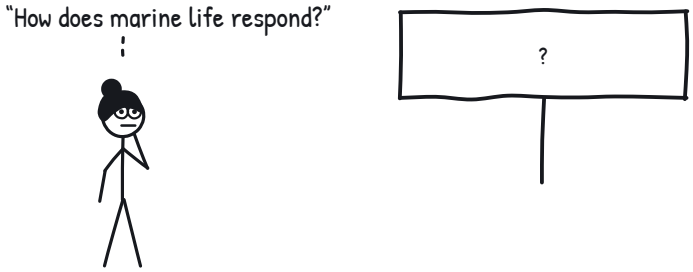
The physiological metrics behind those rates, such as primary productivity, have not fundamentally changed in decades.



The authors think these currencies are unlikely to change in the coming decade, for example when CMIP7 is developed.



Model projections are hindered in part by our inability to untangle how marine life responds to complex ocean change.



Enter omics

Omics studies the genome, transcriptome, proteome and metabolome, with most research on the first three. The authors say these techniques have evolved rapidly this century.



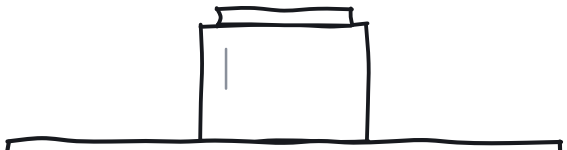
Omics provides a surfeit of data, at a level of detail that is often difficult to relate to the information from physiological rate measurements.



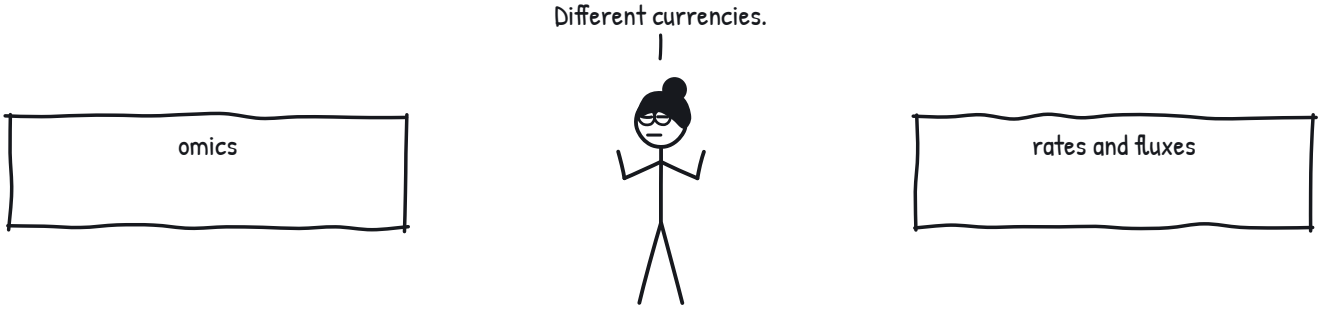
Some people implicitly assume that measuring physiological rates directly is obsolete, because rates can be inferred from omics.



The authors say physiological research has stagnated, in part driven by the advent and rapid evolution of molecular biological techniques.



The authors call the growing mismatch between the currencies of global-scale models (rates and fluxes) and the aspirations of omics something that must be addressed urgently.



The nitrogen test

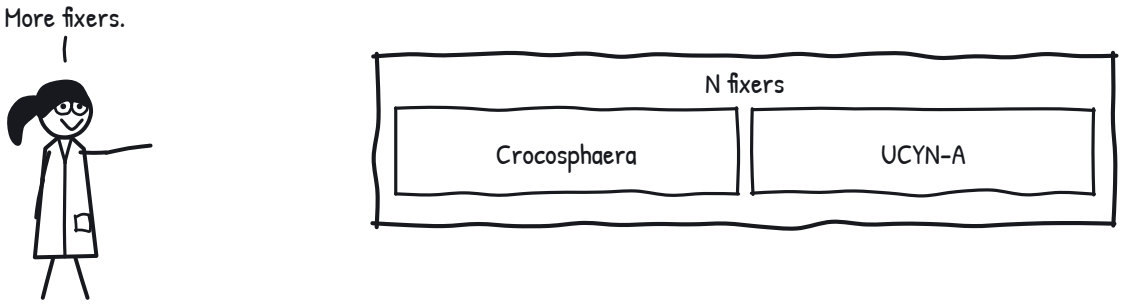
To test the exchange, the authors use three approaches: nitrogen fixation as an example, the recent trajectory of biomedical research, and the ocean phosphorus and iron cycles.



Physiological studies played an early role by quantifying rates of diazotrophy (nitrogen fixation). Those rates were needed to estimate global biogeochemical fluxes of N.



Imbalances in the N fluxes uncovered unidentified N sources, and genetic tools then identified more diazotrophs. Omics has revealed diverse N2 fixers, including the cyanobacteria Crocosphaera and UCYN-A.



But nitrogenase (nif) gene abundance does not directly equate to N2 fixation rates.

nif gene abundance

is not

N2 fixation rate

Genes aren't rates.



Measurements of nif expression are not well correlated with 15N-based rates of N2 fixation.

What about expression?



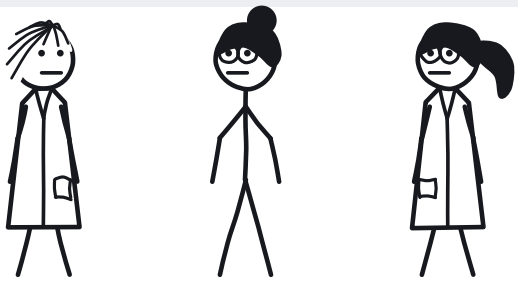


N2 fixation is not included in the CMIP6 models, which currently project declining productivity in low latitude oceans in coming decades.

CMIP6 models

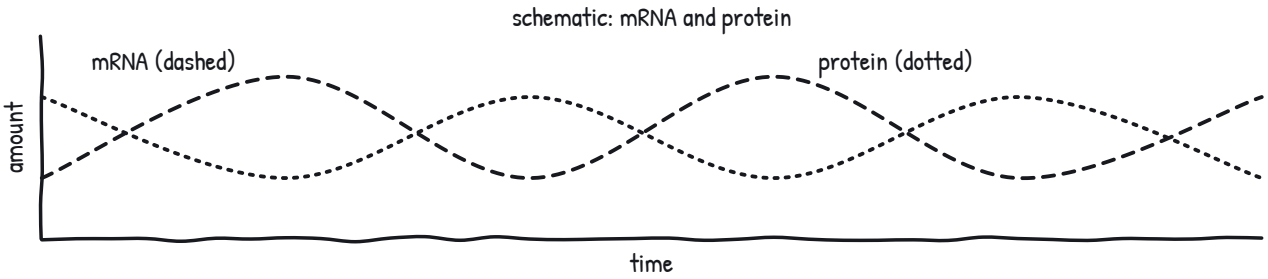
N2 fixation: not included

Both rates and omics will be needed increasingly, say the authors, to reveal and quantify unknown but biogeochemically important pathways for the turnover of N.



Snapshots, not rates

It is well recognized that the amplitude and timing of the mRNA pool does not align with protein expression. A study of one cyanobacteria species, tracked over a day, illustrated the misalignment.



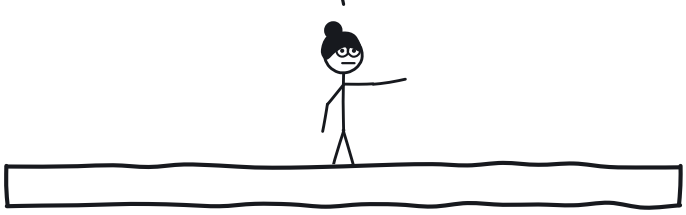
A critique the authors cite says mRNA abundance is a poor proxy for protein abundance.

Noted.

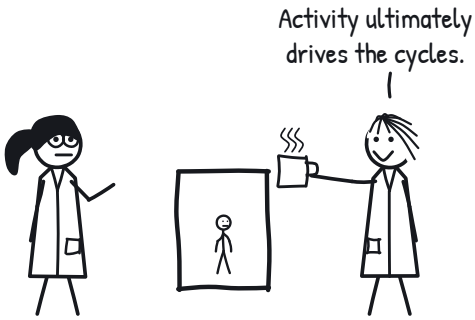


Even in advanced biomedical research, post-transcriptional and post-translational modifications to enzymes confound deriving metabolic rates from omics.

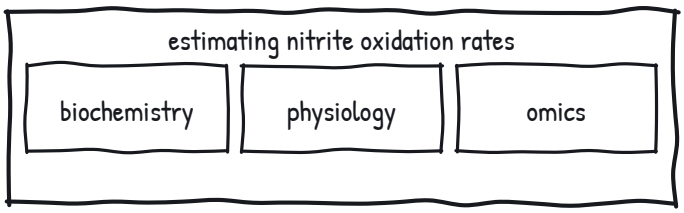
So biomedicine hasn't solved it either.



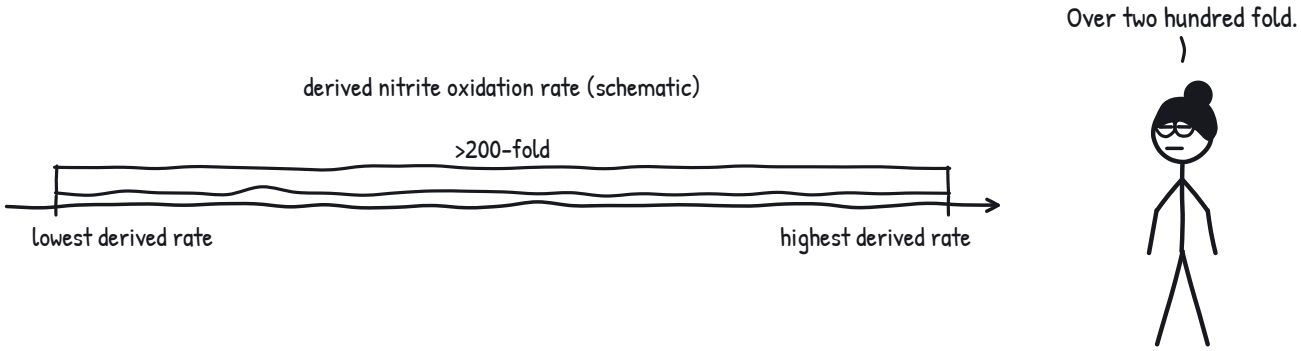
Omics-based approaches provide static snap-shots of physiological potential.



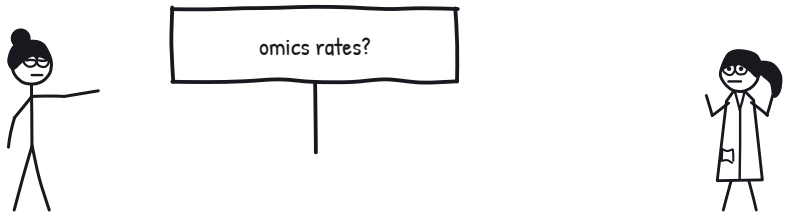
A study of subsurface biota in the Tropical North Pacific found nitrite oxidoreductase of the bacterium Nitrospina abundant. It tried to estimate nitrite oxidation rates from biochemistry, physiology and omics.



Despite this suite of approaches, the derived rates ranged >200-fold, pointing to the need to develop targeted physiological assays.

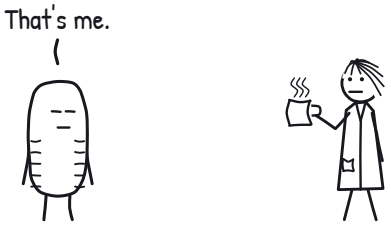


From this and the biomedical record, the authors conclude that it is unlikely that omics can provide such rates in the coming decade.



A currency converter, by accident

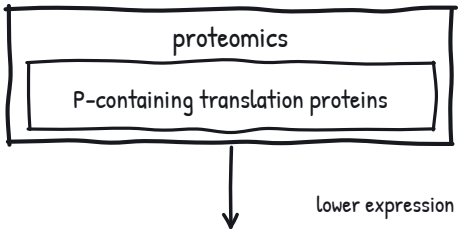
A lab study used proteomics and physiological metrics to explore the cumulative effect of five climate-change stressors on a diatom.



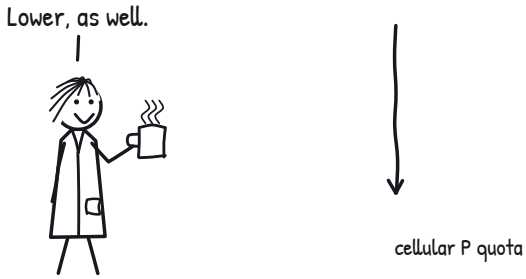
A central finding: the effect of decreased nutrient supply in a future ocean was offset by warming.



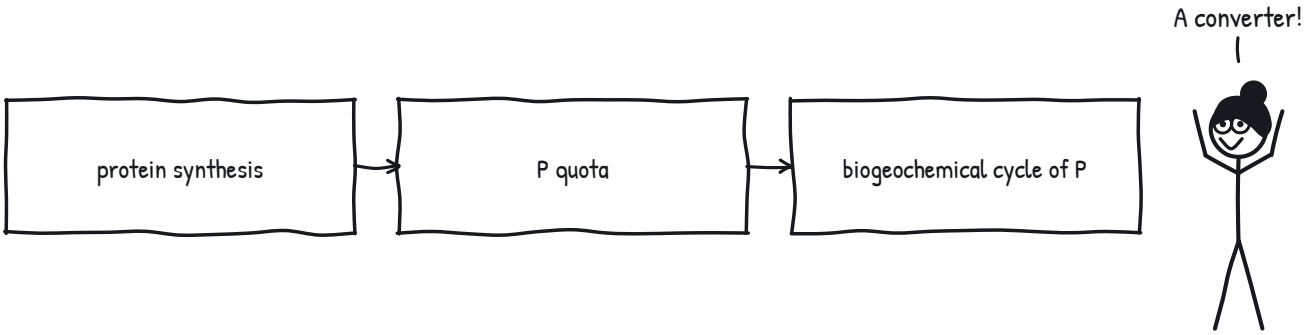
Proteomics revealed that a decreased need for P (phosphorus) was driven by lower expression of P-containing proteins associated with translation.



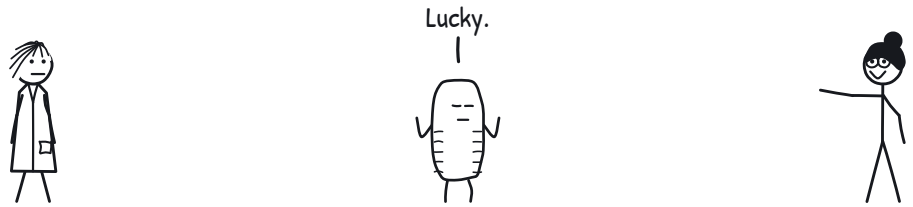
Physiological metrics corroborated this finding: cellular P quotas were lower under warming.



Hence, P quotas acted as a currency converter between protein synthesis and the biogeochemical cycle of P.

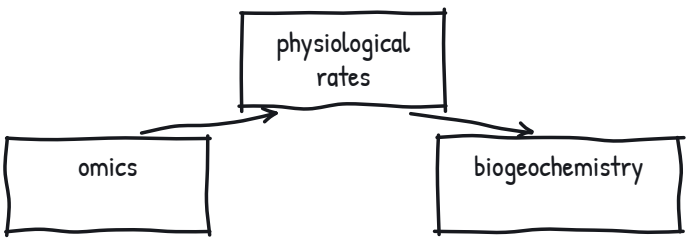


The study showed this link serendipitously. The authors say that in future we must actively seek such conceptual linkages, rather than uncovering them by chance.

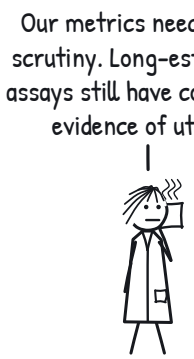


Co-design, and what could go wrong

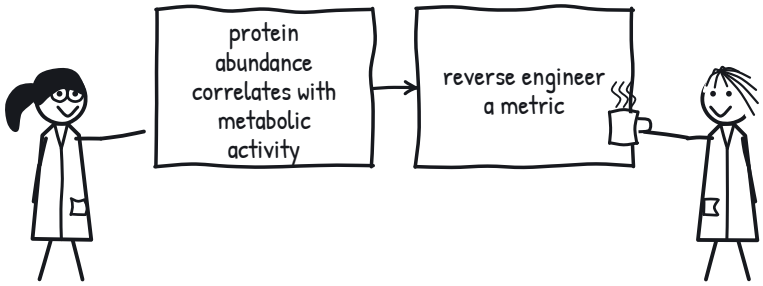
The authors propose that physiological rates can bridge biogeochemistry and omics: they quantify the integrated activity of proteins that drive biogeochemical cycles, in units that modelers can use.



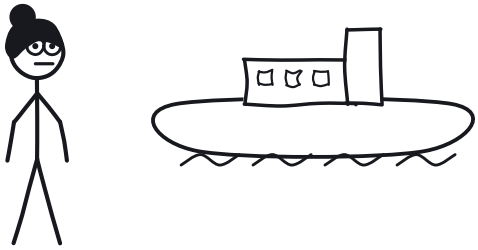
The authors ask whether we are measuring the best physiological metrics to mesh omics with biogeochemistry.



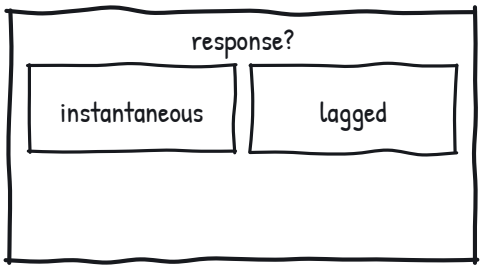
The way ahead may be to use molecular biology to reverse engineer the most pertinent physiological metrics, starting with processes where protein abundance correlates with quantifiable metabolic activity.



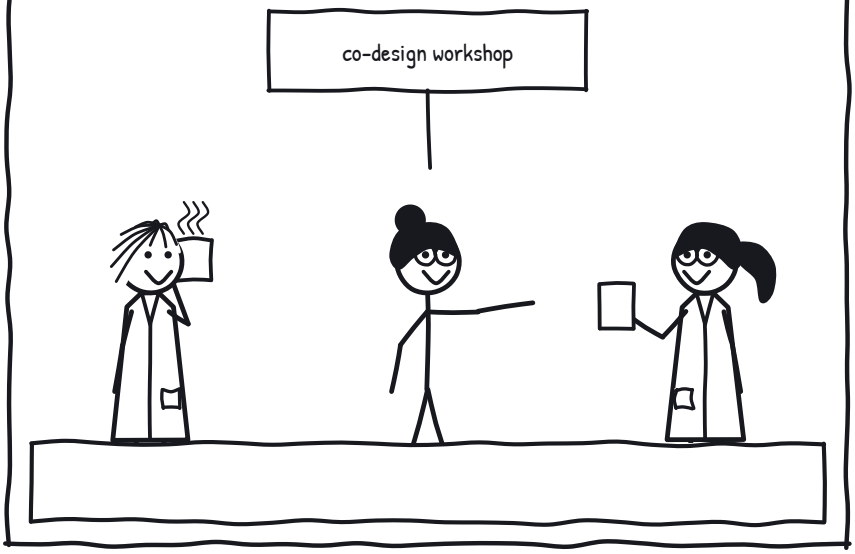
The transition to field studies will face additional challenges, which centre on how marine biota integrate environmental history.



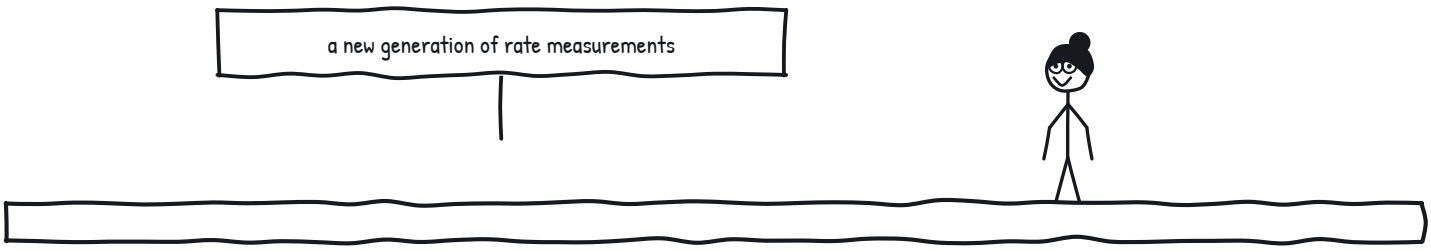
An open question is whether the relationship between environmental forcing and cellular status is instantaneous or lagged. Different metrics, and different omics layers, show a range of response times.



In the authors' opinion, it may be more straight-forward to co-design targeted physiological metrics that better link omics with marine biogeochemistry, helped along by co-design workshops and other forums.



The authors advocate co-designing a new generation of rate measurements. In their opinion, this will lead in the coming decades to incorporating omics into biogeochemical models.



Adapted from "The ongoing need for rates: can physiology and omics come together to co-design the measurements needed to understand complex ocean biogeochemistry?" by Robert F Strzepek, Brook L Nunn, Lennart T Bach, John A Berges, Erica B Young, Philip W Boyd (2022), doi 10.1093/plankt/fbac026. Licence: CC BY 4.0. Condensed into a comic with invented characters and scenes using the science-comic-pipeline; the paper's authors did not write or endorse this document.

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